

SatQuery AI — v4 Progress Report

Expanded multi-dataset fusion model · full plan and current status · SIH26167

GitHub Repo	github.com/sameelkazi/satquery
Companion Doc	v2 Technical Report (published baseline)
Dataset — VRSBench	github.com/xiang709/VRSBench
Dataset — BigEarthNet.txt	arXiv:2603.29630
Dataset — CDVQA	arXiv:2112.06343
Dataset — RSVQA-LR	arXiv:2003.07333
Dataset — Sentinel-1 SAR	zenodo.org/records/10891137

1. Status

PIPELINE COMPLETE · TRAINING IN PROGRESS

v4 is the expanded successor to v2. Every part of the system needed to train it — the data pipeline, the training notebook, the evaluation harness — is built, tested, and running. The model itself is mid-training on Kaggle's free-tier GPUs as of this document. Nothing below is aspirational: it is either already built and running, or explicitly marked as pending.

2. What v4 adds over v2

v2 covers single-image VQA and grounding on one dataset. v4 is trained jointly across five real datasets spanning the full functional scope of the problem statement: single-image VQA and grounding, bi-temporal change-based VQA, and genuine optical–SAR cross-modal fusion — in one coherent fine-tune, not four separate models stitched together.

Dataset	Role	Scale used
VRSBench	Primary single-image VQA, grounding, captioning	up to 120,000 examples
BigEarthNet.txt (arXiv:2603.29630)	Referring-expression / grounding	up to 150,000 examples
CDVQA (arXiv:2112.06343)	Bi-temporal change-based VQA	up to 20,000 examples
RSVQA-LR (arXiv:2003.07333)	Independent VQA benchmark	up to 15,000 examples
BigEarthNet.txt optical + Sentinel-1 SAR	Trained optical–SAR joint reasoning	up to 3,000 examples

3. Architecture

v4 fine-tunes **Qwen3-VL-8B-Instruct** using QLoRA (4-bit NF4 base-weight quantization, rank-64 / alpha-128 LoRA adapters on the language-side attention and MLP projections, vision tower kept frozen), trained via TRL's

SFTTrainer on Kaggle's free-tier GPUs.

4. Engineering already completed

- **Self-contained data pipeline** — every one of the five datasets is fetched and prepared inside the training notebook itself using Kaggle's own internet access. There is no manual download-zip-upload step and no local file paths leaking into a cloud run.
- **Hugging Face Hub caching** — every data-prep script checks for a cached, already-built manifest before re-downloading from source, so re-running or resuming a session does not re-fetch data from scratch.
- **Trained, not scripted, optical–SAR fusion** — each fusion training example pairs BigEarthNet.txt's real optical caption with a second sentence that is a genuinely computed statement derived from that patch's real Sentinel-1 VV/VH backscatter data. The model is being taught to reason jointly over both modalities, rather than having a rule-based comparison bolted on after the fact at inference time.
- **Held-out evaluation harness** — a dedicated evaluation pass scores the trained adapter on genuinely unseen validation data from VRSBench, CDVQA, and RSVQA-LR (confirmed untouched during training), reporting exact-match and token-F1 with saved per-example predictions.
- **Deployment-grade audit trail** — every response the deployed system produces states which model tier answered it, and separates confidence into two explicit categories: derived from real model log-probabilities, or heuristic when no logit signal is available. Nothing is reported as more certain than it actually is.

5. Current status

Training is actively running on Kaggle as of this document. This is a multi-week effort, not a single-session run — it has already gone through several rounds of real debugging against real infrastructure constraints (dependency version conflicts, Kaggle disk-quota limits, dataset caching correctness) rather than a single untested attempt. *[Training duration and current checkpoint to be updated here once available.]*

6. Remaining plan

- Complete the current training run on Kaggle.
- Run the held-out evaluation harness (Section 4) and record exact-match / token-F1 results across all five data sources.
- Validate the optical–SAR fusion outputs qualitatively and quantitatively against real Sentinel-1/2 pairs.
- Publish the trained adapter to Hugging Face Hub and update this document, the GitHub README, and the presentation slide with final, measured numbers.
- Integrate the validated adapter into the backend inference service, replacing the current tiered-fallback default.

7. Known limitations (disclosed upfront)

- Domain gap to the real ISRO Cartosat-2S / RISAT evaluation imagery is a structural risk that no public-data training can fully eliminate — this affects every team working from public data, not just this one. Resolution/scale-jitter augmentation is used to partially mitigate it.
- The optical–SAR fusion training data is newly built specifically for v4 and its quantitative results are not yet available — they will be added to this document once the held-out evaluation (Section 6) is complete.